

CHAPTER 1 - Defining and Collecting Data

SDS188 - Group 3

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1 Defining variables

1.1 Example 1

Four different drinks are sold at a fast food restaurant: fizzy drinks, tea, coffee and bottled water.

1. Explain why the type of drink is an example of a categorical variable.
 - (a) The values can be placed into categories
 - (b) The values arise from a counting process
 - (c) The values represent quantities
 - (d) The values arise from a measuring process
2. Explain why the type of drink is classified according to a nominal scale.
 - (a) The difference between measurements is a meaningful quantity, but the measurements do not have a true zero point.
 - (b) The values are classified into distinct categories in which ranking is implied.
 - (c) The difference between measurements is a meaningful quantity, but the measurements have a true zero point.
 - (d) The values are classified into distinct categories in which ranking is implied.

1.2 Example 2

Businesses are classified according to size: small, medium and large. Explain why the size of businesses is an example of a categorical variable.

1.3 Example 3

The time it takes to download a video is measured.

1. Explain why the download time is a continuous numerical variable.
 - (a) The download time can only be integers.
 - (b) The download time can only be placed into categories.
 - (c) The download time arises from a measuring process.
 - (d) The download time arises from a counting process.
2. Explain why the download time is measured according to a ratio scale.
 - (a) The download time can be placed into categories.
 - (b) The download time does not include true zero point.
 - (c) The numeric values of download time arise from a counting process.
 - (d) The download time includes a true zero point.

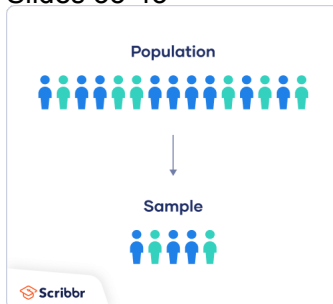
1.4 Example 4

For each of the following variables, determine whether the variable is categorical or numerical. If the variable is numerical, is it discrete or continuous?

1. Number of text messages sent per month
2. Rating (A to F)
3. Time (in minutes) it takes for someone to commute to work
4. Distance (in kilometers) from a person's house to the nearest store
5. Number of televisions in a household

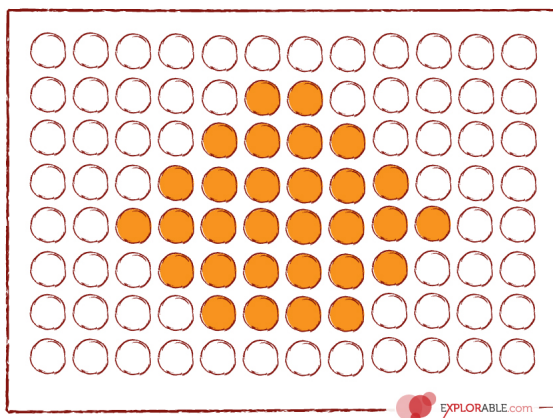
2 Sampling

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2.1 Summary of sampling methods

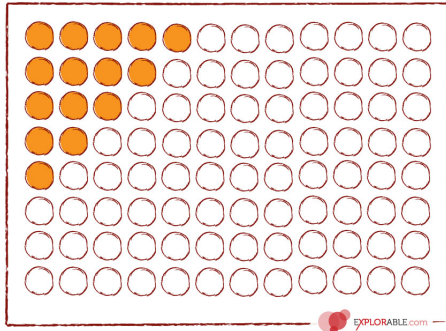
2.1.1 Non-probability sampling



1. Judgment samples

- The sample is purposefully selected to confirm a prior opinion.
- The researcher uses their own judgment to select specific items for their sample.
- For example: A researcher wants to know what is required to graduate with a distinction in university. The only individuals that can advise them about this are people that have already graduated with a distinction.
- Therefore: 'n Judgment sample is used when 'n sample needs to be selected from a specific or limited group of individuals.

2. Convenience samples



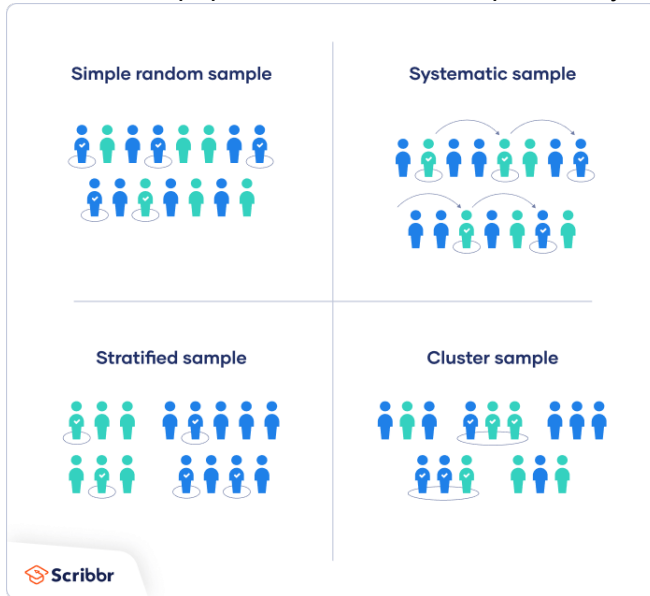
- The items that are selected for the sample consists of those that are easy, cheap and available.
- For example: Select the first five people on the class list or respondents that volunteer to take part in the study.
- Therefore: 'n Convenience sample is selected when the researcher wants to explore basic information or trends at the start of the study.

3. Disadvantages

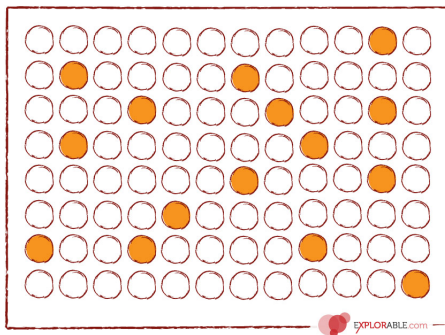
- Leads to biased conclusions, because the sample does not represent the entire sampling frame or population.

2.1.2 Probability sampling

Each item in the population has a known probability of selection.



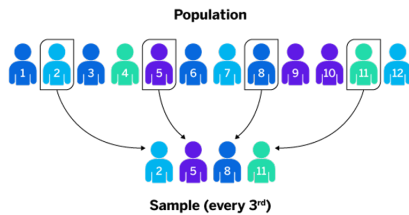
1. Simple Random Sampling



- Simplest form of probability sampling
- Each item in the population has a equal probability of being selected for the sample. The process takes place in a single step where each item is selected independent of the other items in the population.
- Each sample of size n has the same probability of selection, it is considered as fair.
- Leads to a more representative sample

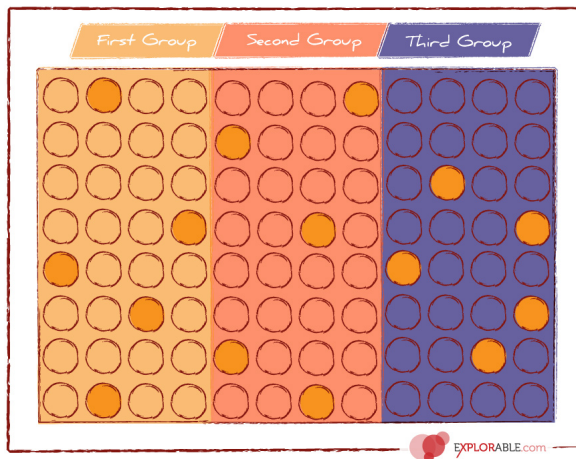
- BUT, 'n complete list of all the items in the population is required

2. Systematic Sampling



- The researcher selects the first item or individual from the population (item with index between 1 and k). The researcher then selects every k'th item in the population.
- The sample is more representative of the population except when certain characteristics of the population are repeated for every n'th item.
- The sample is distributed evenly over the whole population.
- The value k should not correspond with a pattern in the indices of the population.

3. Stratified Sampling



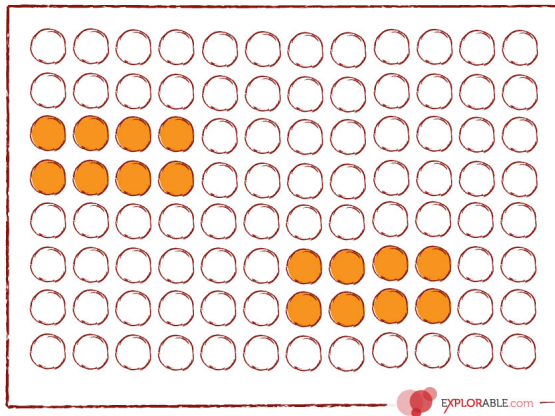
Stratified sampling



- A sampling technique where the researcher divides the population into different subgroups or strata and then selects samples from each strata where the sample sizes are proportional to the sizes of the strata.
- Most common groupings are based on:
 - Age
 - Gender
 - Socio-economic status
 - Nationality
 - Level of education

- Stratified sampling is used when the researcher wants to highlight a specific subgroup in the population. This ensures that the subgroup is represented in the sample.

4. Cluster sampling



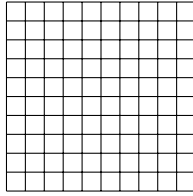
- The researcher first selects groups or clusters and then selects samples (simple random or systematic) from the selected clusters.
- Most common: geographic clusters (countries and cities)

2.2 Compare non-probability and probability sampling

Non-probability sampling	Probability sampling
Samples are selected based on researcher's judgment	Sample is randomly selected
Not all individuals have an equal probability to take part in the study	All of the items in the population have a equal probability of being selected for the sample
The sample is not an accurate representation of the population	Is used when the population consists of different groups
Easy to find respondents	Is used to obtain a more accurate sample
	Not as easy to find respondents

2.3 Sampling with or without replacement

Population ($N = 100$)



Steekproef ($n = 5$)



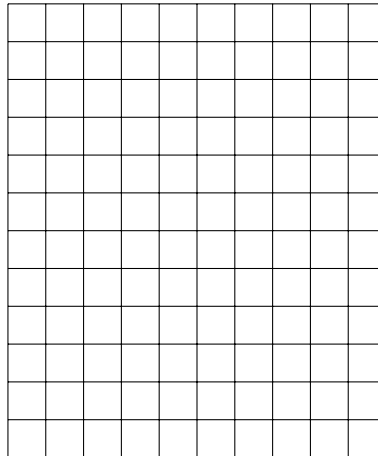
Sampling with replacement:

Each item has a probability of being selected.

Sampling without replacement:

1. Each item has a probability of being selected.
2. Each item has a probability of being selected.

2.4 Examples of systematic sampling



$N = 120$

$n = 10$

$k = 12$

N = 90

n = 18

k =

2.5 Stratified sampling

Select a sample of size n = 2500

Stratum	Individuals in stratum (1000's)	Individuals in sample
Western Cape	6794	290
Eastern Cape	6519	
Northern Cape	1246	
Gauteng	15055	
North West	3997	
KwaZulu Natal	11363	
Free State	2917	
Mpumalanga	4605	
Limpopo	5933	
Total	58429	

$$(N_{stratum}/Total) \times n = \text{individuals in sample per stratum}$$

Round up to nearest integer

For example: $(6794/58429) \times 2500 = 290.6947 \approx 290$

2.6 Comparing stratified and cluster sampling

Stratified	Cluster
Divide population into different groups (e.g. gender or age)	Divide population into comparable groups (e.g. schools or cities)
Select a random sample from each group	Select a random sample from selected groups
Each group's information and characteristics should be available before the sample is selected (expensive)	Samples are selected from only the selected groups (cheaper)

2.7 Examples

- Given a population with $N=940$ individuals, what code number would you assign for:
 - The fifth person on the list?
 - The thirtieth person on the list?
 - The last person on the list?
- For a population of $N=901$, start at row 02 and column 01 of the following table, and determine the number of rows needed to obtain a sample of $n=10$ without replacement.

Row	Column					
	00000	00001	11111	11112	22222	22223
	12345	67890	12345	67890	12345	67890
01	24271	71196	87852	12497	83238	39274
02	36871	50775	30592	57143	17381	68856
03	25853	35041	36871	50775	30592	57143
04	17381	68856	25853	35041	23913	48357
05	63308	16090	51690	54607	72407	55538
06	23913	48357	63308	16090	51690	54607
07	72407	55538	41377	25684	38012	79607
08	94713	97302	57085	86828	33706	05959
09	58947	14870	62976	33199	25798	93525
10	73273	59380	71364	27858	86755	97517

3. Consider a population of $N=93$, start at row 29 of the following table and read from left to right to obtain a sample of $n=15$:

Row	Column							
	00000 12345	00001 67890	11111 12345	11112 67890	22222 12345	22223 67890	33333 12345	33334 67890
28	06873	21440	75593	41373	49502	17972	82578	16364
29	12478	37622	99659	31065	83613	69889	58869	29571
30	57175	55564	65411	42547	70457	03426	72937	83792
31	91616	11075	80103	07831	59309	13276	26710	73000
32	78025	73539	14621	39044	47450	03197	12787	47709

(a) Without replacement

- 12, 47, 83, 76, 22, 65, 93, 10, 61, 36, 89, 58, 86, 92, 71
- 12, 47, 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 99
- 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 89, 58, 86
- 47, 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 89, 58

(b) With replacement

- 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 89, 58, 86
- 12, 47, 83, 76, 22, 65, 93, 19, 65, 83, 61, 36, 89, 58, 86
- 12, 47, 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 89
- 47, 83, 76, 22, 99, 65, 93, 10, 65, 83, 61, 36, 98, 89, 58

4. Suppose that 10 000 clients in a database are categorised according to three types: 3 500 prospective buyers, 4 500 new buyers and 2 000 loyal buyers. A sample of 1 000 clients is requested.

(a) What type of sample should be selected?

- Convenience sample
- Simple random sample
- Systematic sample
- Stratified sample

(b) How would you select this sample?

- Select a simple random sample of each type of client and combine the results

- ii. Divide the N clients in the frame into n groups of k clients, where $k = N/n$, rounded to the nearest integer. Then select the first client randomly from the k clients in the frame. Thereafter, select the remaining $n-1$ clients as each k 'th client.
 - iii. Select the largest sample from the most representative stratum and the smallest sample from the stratum that will most likely have outliers.
 - iv. Select a simple random sample of 500 clients without taking the strata into account.
- (c) Why is this sampling method not simple random?

3 PPS

In certain cases we look at items with associated values, for example invoices with rand values or classes with a number of students. We are interested in selecting a sample of invoices or classes with larger values rather than the items with smaller values. When a simple random sample is selected, the probability of selecting larger values is exactly the same as that of selecting smaller values. The PPS method ensures that the probability of selection is proportional to the size of the items each time an item is selected for the sample.

In short: We choose a sample of n items based on the values (T) associated with each item.

Create a table with 5 columns.

Columns:

1. List of items in the population, e.g. invoices or classes
2. Values associated with each item, e.g. monetary values or number of students
3. Accumulated value. Add the values in Column 2 together, item by item
4. Interval values. The interval's lower limit is the accumulated value of the previous item plus one and the upper limit is the accumulated value for that item
5. Random numbers. Random numbers used for the selection

Steps:

1. Calculate the accumulated value for each item and construct the intervals
2. Use the random numbers and determine in which interval it lies
3. Select the item associated with the interval in which the random number lies

Sampling can be done with or without replacement. Below are two examples that illustrate the process.

3.1 Example 1: Invoices and monetary values

A population consists of 1500 invoices with a total monetary value of R225 000. Select a sample of 20 invoices.

Random numbers: 000098, 000517, 000972, 006863

Invoice	Monetary value (R)	Accumulated value (R)	Interval values	Random numbers
1	10	10	000001-000010	
2 *	112	10+112=122	000011-000122	000098
3	78	122+78=200	000123-000200	
4	150	200+150=350	000201-000350	
5 * *	5872	350+5872=6222	000351-006222	000517 000972
6	613	6222+613=6835	006223-006835	
7 *	114	6835+114=6949	006836-006949	006863
8	14	6949+14=6963	006950-006963	
...	...	...	...	
		225000		

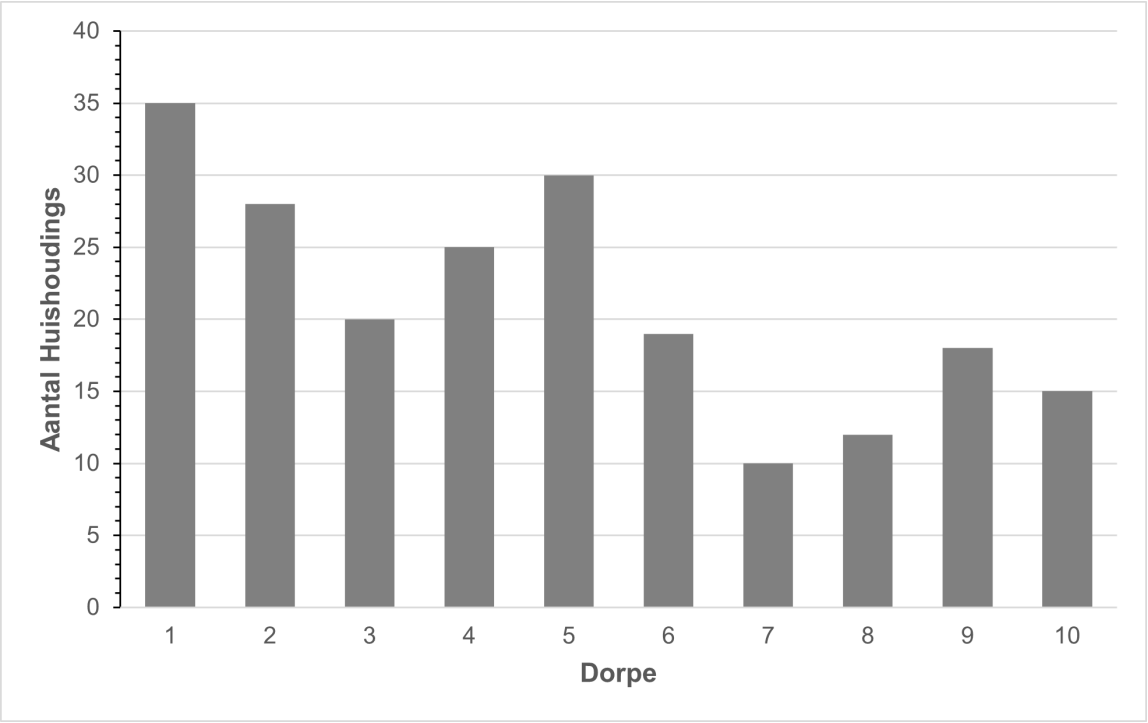
The first 4 items in the sample are thus obtained as invoices 2, 5, 5 and 7. This sample is selected with replacement, which means that invoices can be selected for the sample more than once.

3.2 Example 2: Towns and number of households

A population consists of ten towns with a total of 212 households. Select a sample of six towns by using the PPS method.

Town	1	2	3	4	5	6	7	8	9	10
Number of households	35	28	20	25	30	19	10	12	18	15

The bar graph below illustrates the sized of the values for each item. The PPS method ensures that the towns with less households have a smaller probability of being selected for the sample.



3.2.1 Sampling with replacement

Random numbers: 181, 059, 201, 073, 071, 066

Town	Number of households	Accumulated number	Interval values	Random numbers
1	35			
2	28			
3	20			
4	25			
5	30			
6	19			
7	10			
8	12			
9	18			
10	15			

The sample consists of the following towns:

3.2.2 Sampling without replacement

Random numbers: 130, 143, 019, 047, 007, 105, 226, 125, 154, 190

Town	Number of households	Accumulated number	Interval values	Random numbers
1	35			
2	28			
3	20			
4	25			
5	30			
6	19			
7	10			
8	12			
9	18			
10	15			

The sample consists of the following towns:

4 Formatting data

4.1 Identify irregularities and missing values in the data

A survey is conducted regarding which mobile phone brand each respondent owns. The options include:

- Alpha
- Beta
- Gamma
- Delta
- Epsilon
- Zeta
- Mu

The brands of the 20 respondents in the sample are given in the table below

Respondent	Brand of mobile phone
1	Mu
2	Moo
3	Alpha
4	Epsilon
5	Beta
6	Gamma
7	Mu
8	Alpha
9	
10	Gamma
11	Beat
12	Delta
13	Alpha
14	MU
15	Alpha
16	Zeta
17	Mu
18	Mu
19	Aleph
20	Alpha

Which irregularities can you identify? Choose all applicable options (can be more than one).

☐	The 14th value
☐	The 5th value
☐	The 11th value
☐	The 19th value
☐	The 17th value
☐	The 2nd value
☐	The 3rd value
☐	The 6th value

Are there any missing values in the dataset?

4.2 Reorganise the data

A hotel company manages ten hotels in a holiday resort. The hotels have rooms in different price classes: cheap, average and expensive. Data is collected that indicates the number of rooms in use every day for a month. How can this data be recoded into three price class categories? Choose the correct option:

1. Create three new numerical variables: number of cheap rooms in use, number of average price rooms in use and number of expensive rooms in use.
2. Create three new categorical variables, one for each price class. Determine how many hotels include cheap rooms and allocate all the data for those hotels to a new categorical variable: cheap hotel rooms. Determine how many hotels include average price rooms and allocate all the data for those hotels to a new categorical variable: average price hotel rooms. Determine how many hotels include expensive rooms and allocate all the data for those hotels to a new categorical variable: expensive hotel rooms.